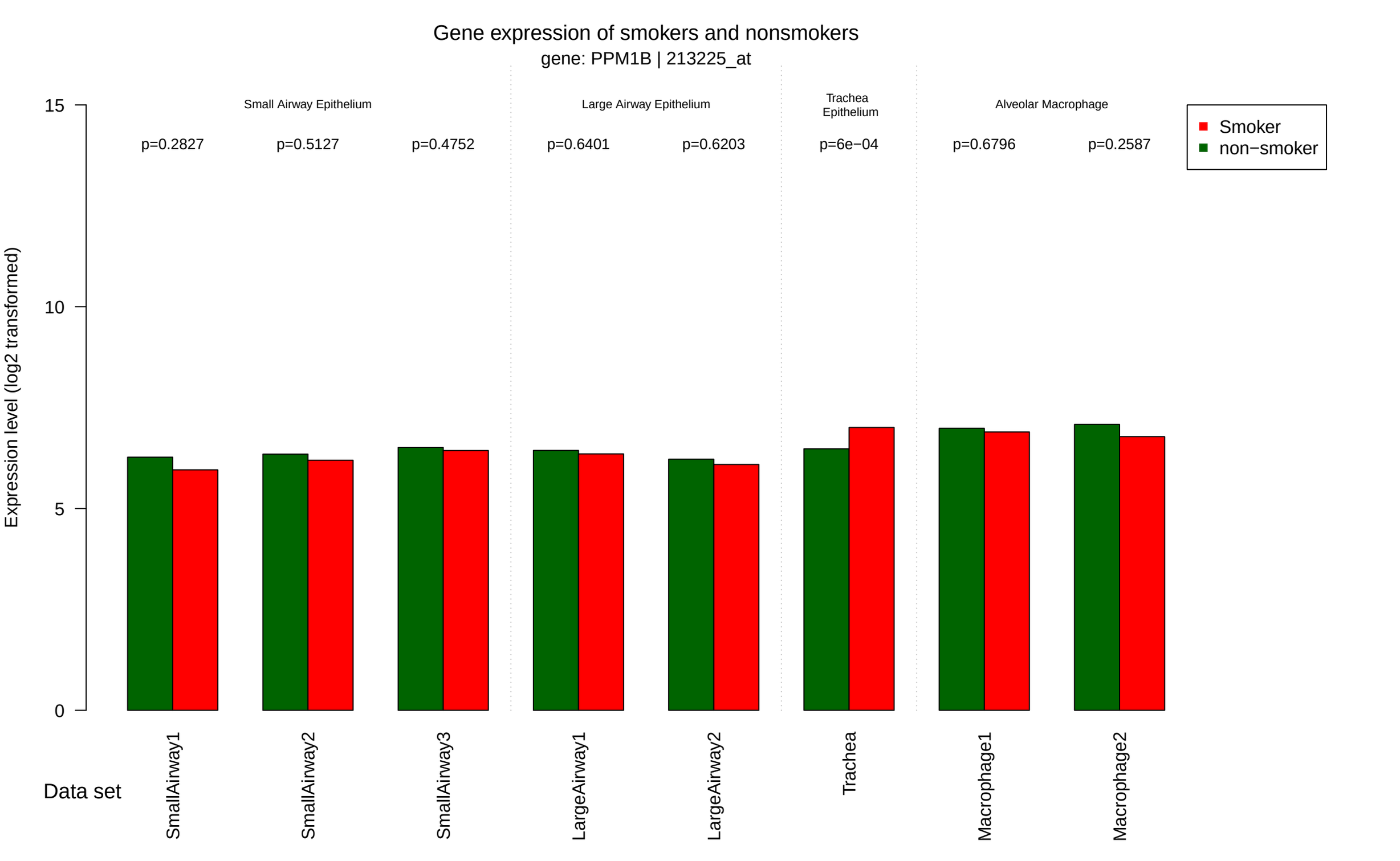
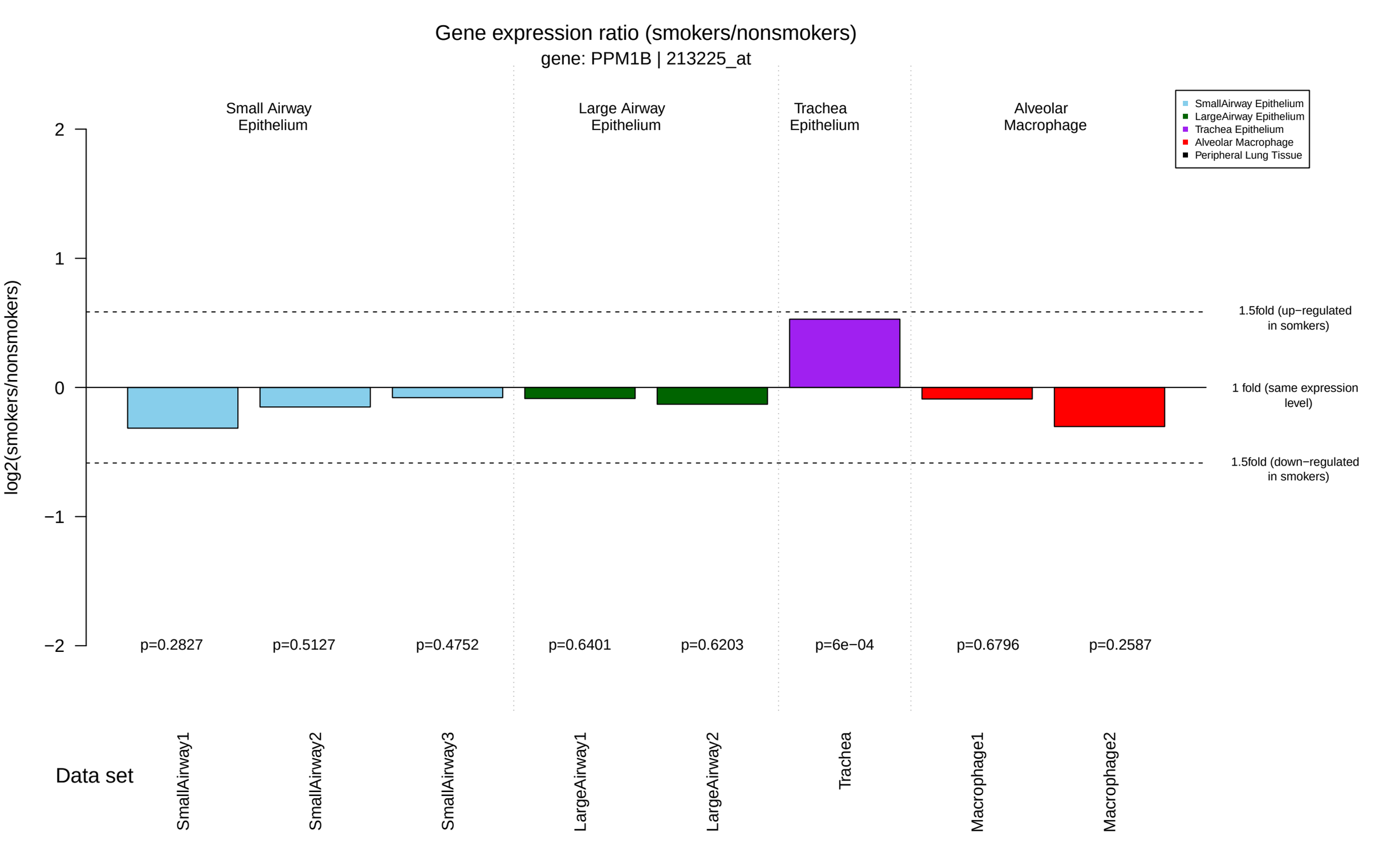
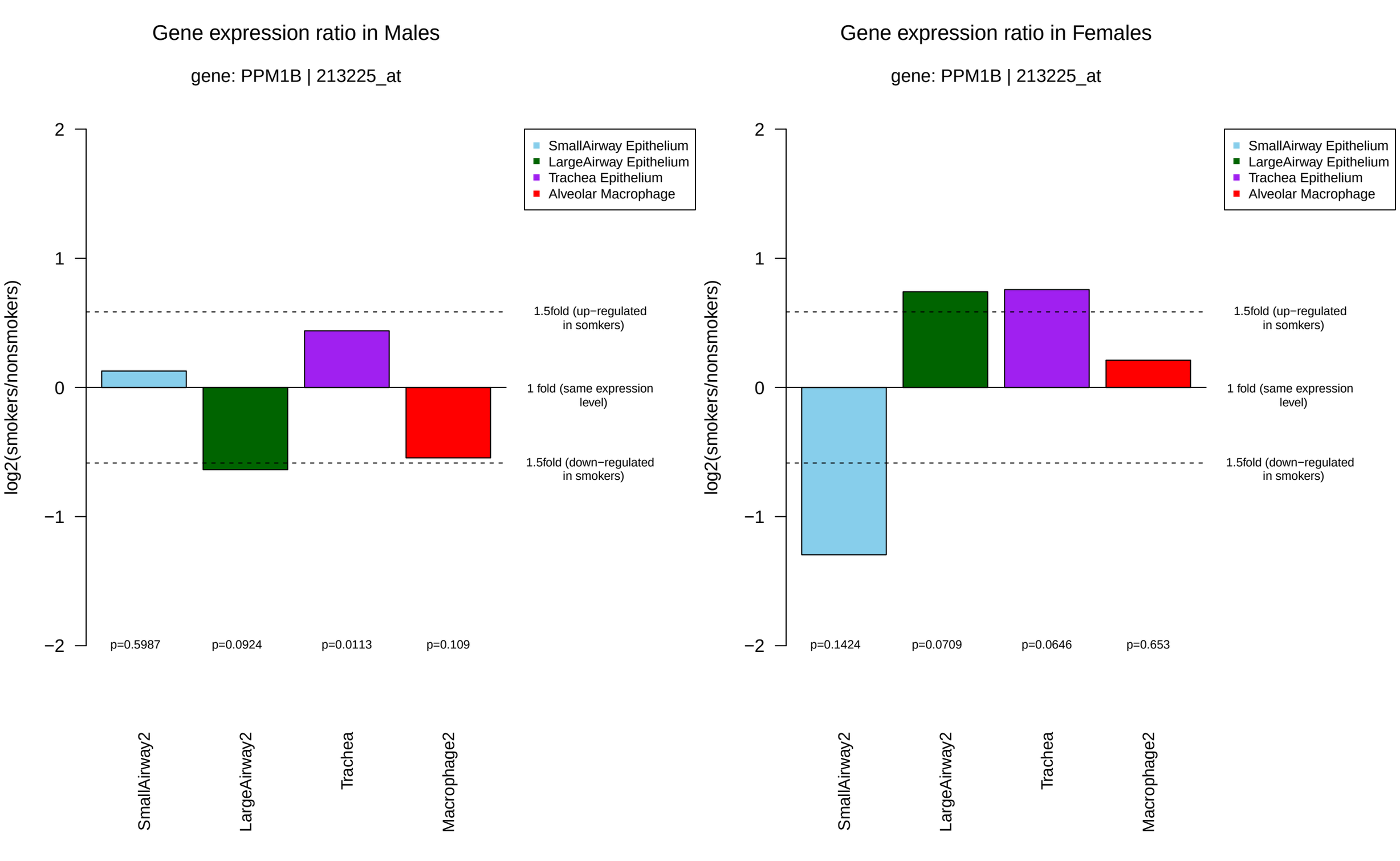
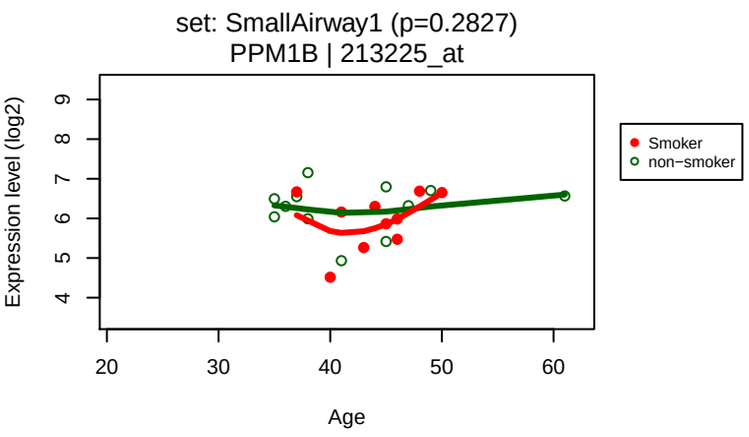
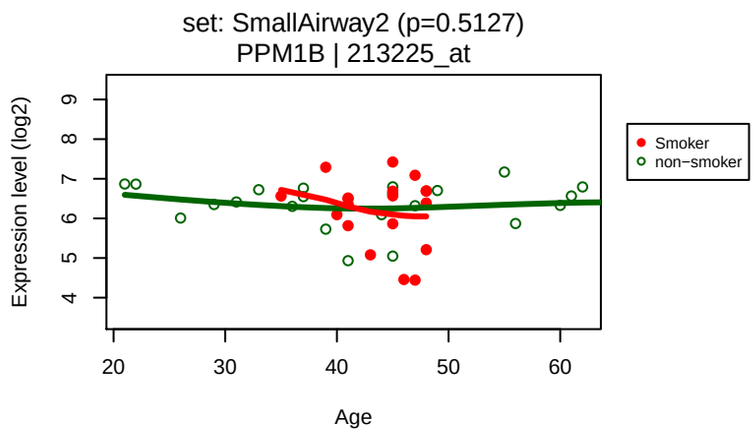
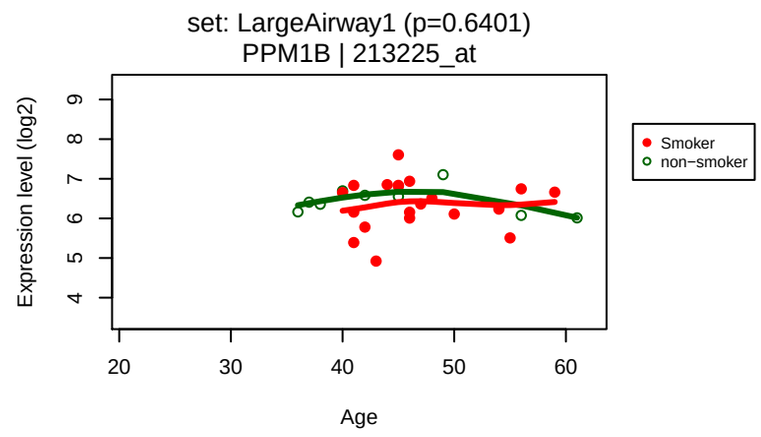

Gene expression of smokers and nonsmokers
gene: PPM1B | 213225_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2827
p=0.5127
p=0.4752
p=0.6401
p=0.6203
p=6e−04
p=0.6796
p=0.2587
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

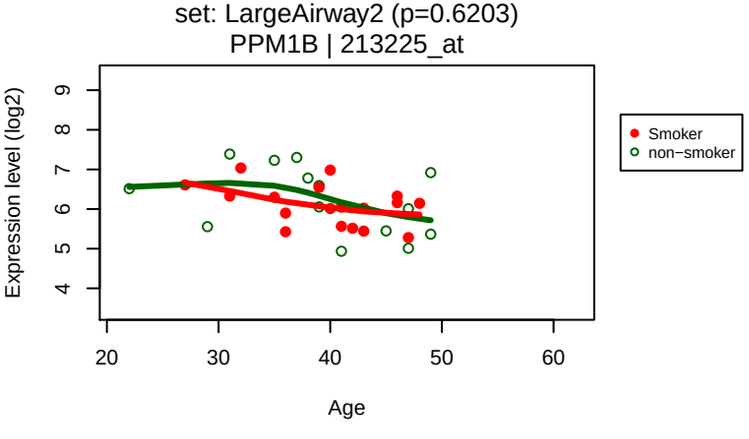

Gene expression ratio (smokers/nonsmokers)
gene: PPM1B | 213225_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2827
p=0.5127
p=0.4752
p=0.6401
p=0.6203
p=6e−04
p=0.6796
p=0.2587
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

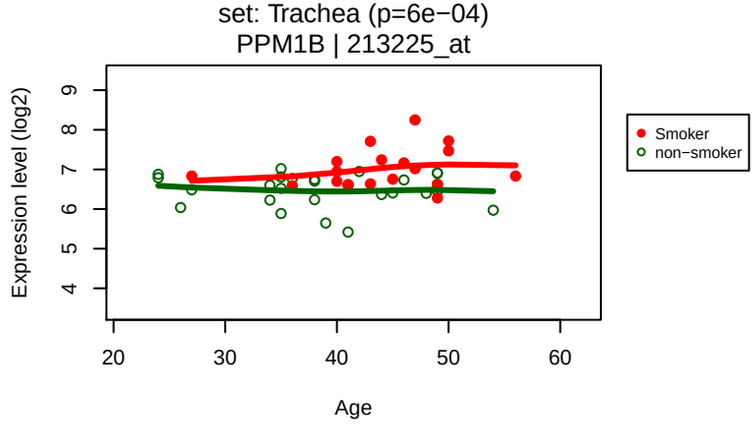

Gene expression ratio in Males
gene: PPM1B | 213225_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5987
p=0.0924
p=0.0113
p=0.109
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PPM1B | 213225_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1424
p=0.0709
p=0.0646
p=0.653
SmallAirway2
LargeAirway2
Trachea
Macrophage2

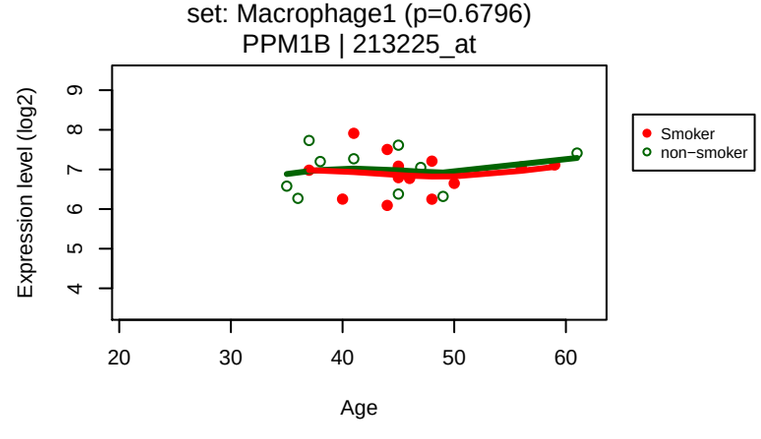

set: SmallAirway1 (p=0.2827)
PPM1B | 213225_at
Expression level (log2)
Age
Smoker
non−smoker

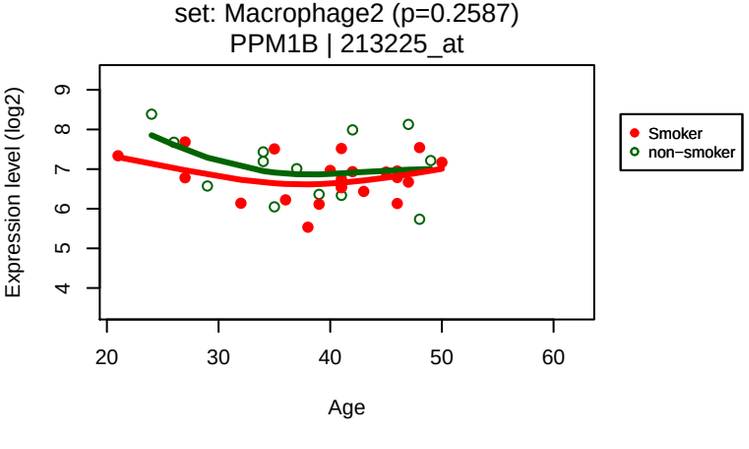

set: SmallAirway2 (p=0.5127)
PPM1B | 213225_at
Expression level (log2)
Age
Smoker
non−smoker

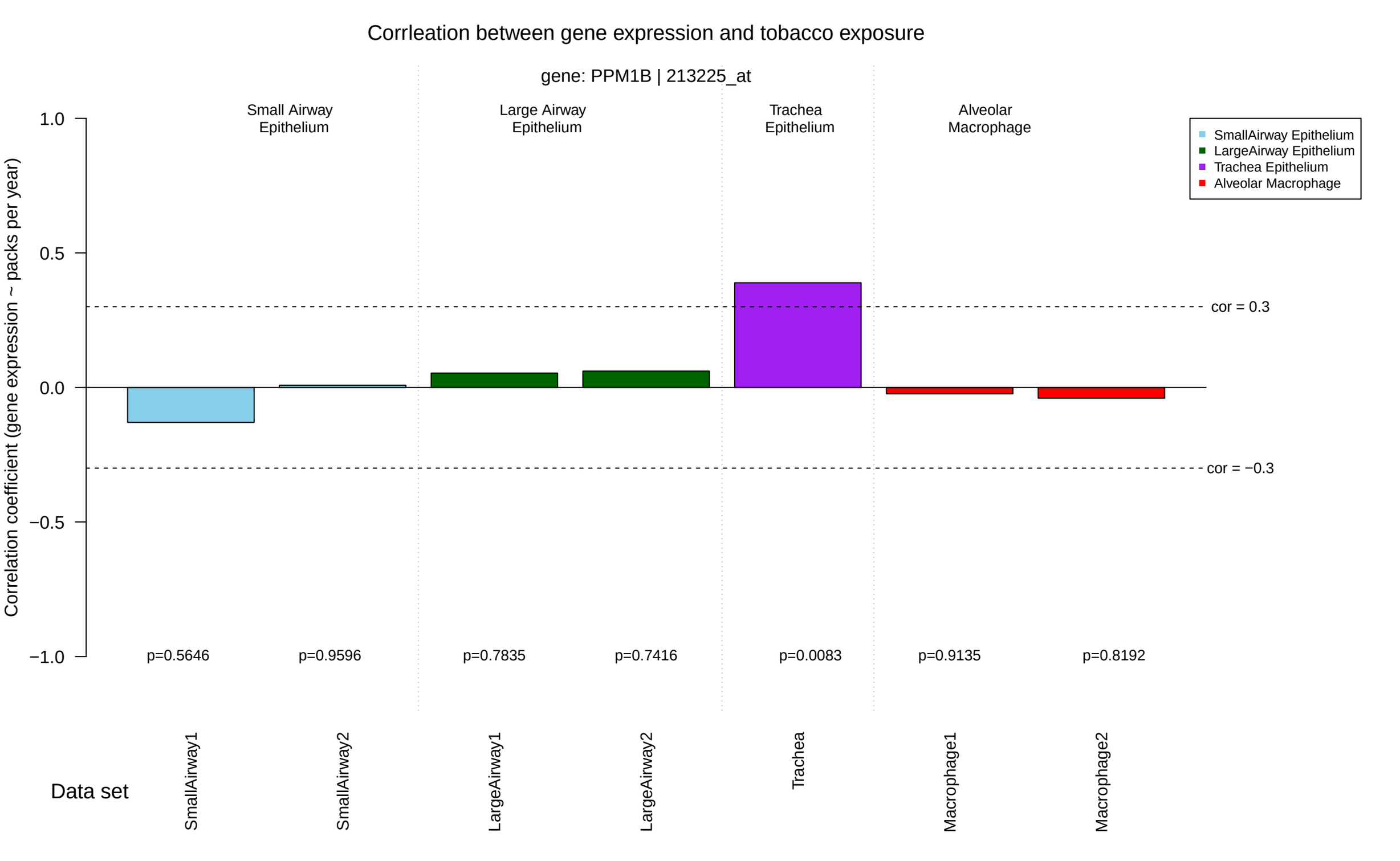

set: LargeAirway1 (p=0.6401)
PPM1B | 213225_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6203)
PPM1B | 213225_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=6e−04)
PPM1B | 213225_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6796)
PPM1B | 213225_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2587)
PPM1B | 213225_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PPM1B | 213225_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.5646
p=0.9596
p=0.7835
p=0.7416
p=0.0083
p=0.9135
p=0.8192
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2